

Routledge
Taylor & Francis Group

Defence and Peace Economics

ISSN: 1024-2694 (Print) 1476-8267 (Online) Journal homepage: www.tandfonline.com/journals/gdpe20

DEFENCE AND PEACE ECONOMICS: THE SECOND DECADE IN RETROSPECT

Daniel Arce & Christos Kollias

To cite this article: Daniel Arce & Christos Kollias (2010) DEFENCE AND PEACE ECONOMICS: THE SECOND DECADE IN RETROSPECT, Defence and Peace Economics, 21:5-6, 395-407, DOI: 10.1080/10242694.2010.512195

To link to this article: https://doi.org/10.1080/10242694.2010.512195

Published online: 14 Dec 2010.

Submit your article to this journal

Article views: 300

View related articles

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=gdpe20

Defence and Peace Economics, 2010
Vol. 21(5–6), October–December, pp. 395–407

Routledge
Taylor & Francis Group

DEFENCE AND PEACE ECONOMICS: THE SECOND DECADE IN RETROSPECT

DANIEL ARCE[a] AND CHRISTOS KOLLIAS[b]*

[a]*School of Political and Policy Sciences, University of Texas at Dallas, USA;* [b]*Department of Economics, University of Thessaly, Greece*

(Received in final form 27 July 2010)

INTRODUCTION

Twenty years since its inception and original publication in 1990, *Defence and Peace Economics* is now a well-established, reputable, international academic journal. As a testament to its success, DPE has recently been included in the field of Economics and Business in *Essential Science Indicators* from Thomson Reuters. Although DPE covers a specialised subject area within the discipline of economics, the importance of new and more traditional security issues is such that both the readership as well as the coverage of DPE, has steadily grown over the years. Subscribers are distributed worldwide and include universities, government agencies, research institutions, and think tanks. Similarly, the topics covered in the journal have also grown and shifted, reflecting new security and conflict-related issues and challenges that have emerged in the post-bipolar and post 9/11 world. Indeed, from the seminal contribution half a century ago of Hitch and McKean (1960), *The Economics of Defense in the Nuclear Age*, the field of defence and security economics has steadily expanded, as have the number of defence and security economists worldwide, with many important, challenging and cutting edge contributions having appeared in DPE.

The basic scope of the journal remains the same. It aims to attract high quality research on all aspects of the economics of defence, security, disarmament and peace broadly defined. In the last 20 years, it has attracted the research of scholars around the world on themes such as the economic effects of defence spending; terrorism; the defence-industrial complex; defence R&D and technology; arms races; conflict and civil war; peace and peacekeeping; the arms trade; the economics of alliances and burden sharing; defence management, budgeting and procurement; military personnel and recruitment; country studies and many other issues. Contributions were both of an applied as well as a theoretical nature that relied on the methodology and tools from many other fields and disciplines such as micro and macroeconomics, public economics, public choice, game theory, regional economics, general and partial

*Corresponding author. Email: kollias@uth.gr

ISSN 1024-2694 print: ISSN 1476-8267 online © 2010 Taylor & Francis
DOI: 10.1080/10242694.2010.512195

equilibrium analysis, industrial economics and organisation, labour economics, political science and others.

This article offers a retrospective view of the second decade of DPE. It summarises the contributions and coverage over the last ten years (i.e. volumes 11–20) and compares the thematic focus to that of the first ten volumes. The significant changes observed reflect important shifts in global defence and security challenges and their evolution throughout the years. The paper also subjectively points to gaps and possible new directions of research over the next decade. Clearly, the choice of topics reflects the authors' views and analysis and should by no means be regarded as exhaustive. It simply aims to highlight some of the many possible research directions for the years ahead and to stimulate the discussion and exchange of ideas among defence and security economists in academia and elsewhere.

COVERAGE AND THEMATIC FOCUS IN THE SECOND DECADE

Well over 300 articles, both of a theoretical and empirical nature, have been published during the second decade of DPE. Compared with the 207 papers (207) of the first ten volumes of the journal, the 331 papers in volumes 11 to 20 represent a significant increase of around 60% (Figure 1). This increase reflects growing interest in the broader field of defence, peace and security economics and, as a consequence, the rise in the number of papers submitted to the journal. As a reminder, it is worth mentioning that this growing interest also caused the journal's publication to increase, from volume 12, in 2001, to six issues per year from the original four issues per volume when it was originally launched in 1990, by the founding editors Keith Hartley and Todd Sandler.

Keeping with the tradition of the 10-year retrospective analysis of Hartley and Sandler (2000), the next step was to try to group the published articles in thematic categories. As mentioned above, the coverage of the journal includes topics ranging from the economic effects of defence spending to defence management; from the defence industrial base to terrorism; from the economics of alliances to conflict and civil war; from military personnel and recruitment to defence R&D and technology; from arms races to issues of peace and peace keeping. The thematic grouping of published papers is useful in a number ways. Apart from revealing the primary foci of research in the broader field of defence, peace and security economics; the grouping in thematic categories also helps to track changes and underlying currents in research vis-à-vis the first ten volumes of the journal. Indeed, such shifts in the

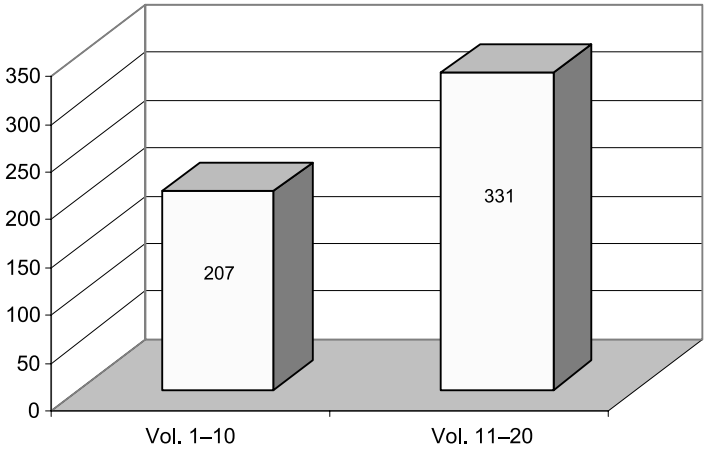


FIGURE 1 Number of articles published in volumes 1–10 and 11–20

research agenda – as will be shown further below – can be directly associated with the major changes in the international security environment and the concomitant issues that have emerged as new security challenges.

To group more than 300 papers is somewhat challenging. As was also acknowledged in the case of the ten-year retrospective paper by Hartley and Sandler (2000), the problem with such summaries is twofold. First, one has to decide what the broad categories will be. Second, articles have to be categorized. The number of categories used to group the papers is an important issue since if they are too few then the level of aggregation is such that we may lose important information. On the other hand, if too many groups are used then it is possible that important information gets blurred and does not stand out. Clearly, it is difficult to find the optimum number of thematic groups in an objective manner.

The next question that arose was what groups to create. The choice of the broad thematic categories of coverage proved a bit more difficult to negotiate, although a significant number of groups were self-evident. Ideally, the choice would have been to stick to the categories of the ten-year retrospective paper. Although this would have better served the need for comparison, it was felt that it would misrepresent significant and noteworthy changes that have occurred in the thematic coverage and focus over the last ten years. Thus, although most of the thematic categories of the first decade retrospective summary were kept as they were, albeit some of them slightly renamed, some were broken up into two while others were grouped together. Hopefully a balance was achieved by the decisions made that serves both the need of comparability between the two decades as well as allowing for the important changes that have occurred to come to light.

The final task was that of assigning papers to categories. For some it was easy and self evident. Some papers deal with more than a single issue and hence fall in more than one category or sub-area of defence economics. Thus, the papers of volumes 11–20 were placed in the various groups on the basis of what was deemed to be their primary focus. Just as in the case of the Hartley and Sandler (2000), this was a subjective decision since we had to exercise our judgement and our decisions may not necessarily coincide with what the author(s) would have decided as the proper thematic category in which to place their paper. Undoubtedly, some of the authors may disagree with our decisions but it would have been a tremendously difficult and time consuming task to contact all the authors and ask them to decide on this. We can only hope that we have not grossly misplaced their articles.

Given the issues discussed above, 16 groups were created. This represents slightly fewer compared with the 21 groups of the Hartley and Sandler (2000) first decade summary. The papers to be categorized were of a theoretical, empirical, conceptual or policy-oriented nature. A plethora of concepts, tools and methodologies has been used, covering almost all major disciplines of economics and, of course, virtually every econometric technique. The groups as well as the number of papers within each group are presented in Table I. Here, it should be stressed that the categories are defined in the broadest possible terms. For example, in the category *alliances*, papers that deal with issues such as burden sharing in the context of the common European defence policy have been included, although an EU military alliance is by no means in existence. Likewise, in the *defence and the economy* group, all papers that deal with the many issues associated with the economic impact and effects of defence expenditures are included. Indeed, as can be seen in the relevant table, over the last ten years, this is the dominant category that has attracted most articles. Similarly, in *defence policies, planning and management*, all papers that deal with issues such as procurement, defence budgeting and budget allocation, military policy and posture, etc, are included.

Let us now turn to the changes in coverage that have occurred between the first and second decade. In this context, there are some clear winners and losers that emerged in terms of the shift in the thematic focus that has taken place. This shift in coverage is largely attributable to

TABLE I Thematic Coverage in Volumes 11–20

Thematic groups	*Number of articles*
Defence and the economy	56
Terrorism	37
Defence policies, planning & management	30
Conflict & civil wars	29
Defence industries and industrial base	25
Peace and peacekeeping	23
Arms races, arms control and disarmament	22
Economics of alliances	20
Defence R&D and technology	17
Fiscal effects of defence spending	16
Military personnel & recruitment	14
Models of defence expenditures	12
Arms trade	10
Country surveys	9
Others	7
Regional economics and defence	4
Total	331
Book reviews	14

the changes that have taken place, sometimes in a dramatic fashion, as in the case of 9/11, in the international security environment and agenda. The most dramatic representation of this is the number of papers that address issues of *terrorism* (37) and *conflict & civil wars* (29). They are the second and fourth most popular categories in volumes 11–20. The corresponding group in the 10-year retrospective analysis of Hartley and Sandler (2000) was *terrorism, guerrilla warfare and insurgencies*. It attracted only 11 papers. Clearly, this significant change can be directly attributed to the dramatic events of 9/11 as well as to other major terrorist incidents worldwide, such as the Madrid and London terrorist attacks in 2004 and 2005 respectively (Enders and Sandler, 2006). Such important events signify the emergence of transnational terrorism as a predominant and global security concern at the turn of the century. Naturally, the multitude of issues around terrorism have attracted and continue to attract the growing attention of researchers, with contributions of both a theoretical and applied nature covering a wide spectrum of terrorist-related issues. It is such noteworthy changes in the coverage that very much imposed the need to alter the thematic categories in which papers were grouped.

The various economic effects of defence spending, as represented in *defence and the economy* (a total of 56 papers), is by far the category that appears to have increased its coverage exponentially. This is more so if the 16 papers in *fiscal effects of defence spending* are also included in a broader group. *Terrorism*, with 37 papers also clearly emerges as major winner if a 'then and now' perspective, for reasons of comparison, is adopted. Similarly for *conflict and civil wars*, with 29 papers. *Arms races, arms control and disarmament* also emerge as a winner with 22 papers, whereas the corresponding groups in the 10-year retrospective summary *arms races* (6) and *disarmament and arms control* (5) had attracted roughly half the papers. Others have remained pretty much unchanged; the *economics of alliances* with 20 papers compared with eighteen in volumes 1–10; *military personnel and recruitment* (14) vis-à-vis *military manpower* (14); *defence policies, planning and management* (30) vis-à-vis the three corresponding groups in volumes 1–10, i.e. *procurement processes and policies* (21), *military posture and strategic policy* (6) and *defence budgets* (3). Obviously, for a statistically proper and accurate picture to emerge, these absolute numbers of papers have to

TABLE II Coverage Comparison between Volumes 1–10 and 11–20

Volumes 11–20		*Volumes 1–10*	
Thematic category	*No*	*Thematic category*	*No*
Defence and the Economy	56	Defence & growth	9
		Developing countries and defence	2
Terrorism	37	Terrorism, guerrilla warfare & insurgencies	11
Conflict & Civil Wars	29		
Defence Policies, Planning & Management	30	Procurement processes & policies	21
		Military posture & strategic policy	6
		Defence budgets	3
Defence Industries and Industrial Base	25	Defence industries & industrial base	16
		Conversion	17
Peace and Peacekeeping	23		
Arms Races, Arms Control and Disarmament	22	Arms races	6
		Disarmament & arms control	5
Economics of Alliances	20	Economics of alliances	18
Defence R&D and Technology	17	Defence R&D, technological change & weapon development	6
Fiscal Effects of Defence Spending	16		
Military Personnel & Recruitment	14	Military manpower	14
Models of Defence Expenditures	12	Defence expenditure studies	29
		Military production function	4
Arms Trade	10	Arms trade	6
Country Surveys	9	Country surveys	11
Others	7	Others	5
Regional Economics and Defence	4	Regional economics & defence	7
		Public choice & defence	8
		Nature of defence economics	4

be weighted with the total number of published articles that have increased by about 60% in the last ten volumes, as already seen. For comparison purposes, in Table II the number of papers that each thematic group attracted, are presented. For each thematic category of volumes 11–20 we have placed next to it the group (or groups) of the 10-year retrospective summary that, in broad terms, can be treated as the corresponding categories with the exception of a couple of groups for which no clear corresponding category exists.

Apart from the observed changes in the coverage that have been highlighted, a noteworthy slight change in the concentration of the published articles in the top thematic groups has also been the case. As shown in Figure 2, the number of papers that the first ten thematic categories attracted (275) rose to just above 83% of the total papers published in the second decade compared with 74.4% of the first ten volumes. This also appears to be the case for the first three groups – just over 37% from 33% – and the first five groups, the share of which increased to 53.5% from 48.8% (Figure 2).

Perhaps worth mentioning here is the share that the ‘champion’ categories of the journal’s coverage had in the total number of articles in ECONLIT (Table III). Clearly, this share reflects the journal’s importance and standing within this field of economic analysis; something that one hopes will increase and improve further in the years to come. Of the total number of papers in ECONLIT that deal with the economic effects of defence expenditure 52.5% were published in DPE. This share is even higher when it comes to the economic

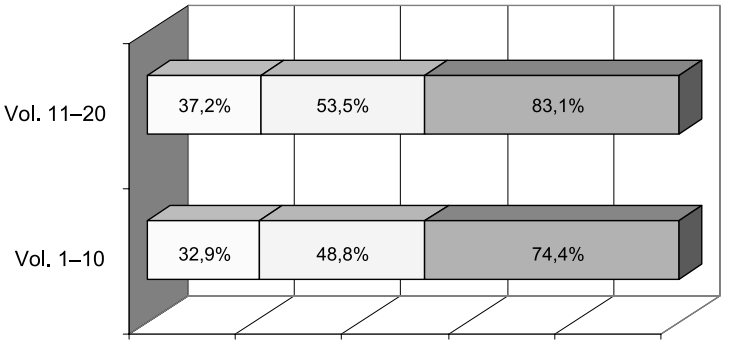


FIGURE 2 Concentration in the first three, five and ten groups

TABLE III DPE's 'Champion' Categories in Volumes 11–20 as a Percentage of ECONLIT

Category	*% of ECONLIT Journal articles in DPE*
Economic effects of defence expenditures	52.5
Terrorism	6.9
Civil war	3.8
Defence industry	23.9
Arms race/disarmament	56.3
Defence research & development	22.2
Fiscal effect of defence	32.9
Economic analysis of alliances	60.4

analysis of alliances, with 60.4% of all the papers published in our journal; and 56.3% of papers in the thematic category arms race/disarmament. Lower are the percentage shares when it comes to other categories such as defence industry, terrorism, civil war. We feel that in the next decade DPE should aspire to attract a greater share in some of these categories.

One of the distinctive features of DPE is the country survey series, with a standing call for papers, that focus on surveying the defence sector and policies of different countries. The aim is to publish papers that offer a comprehensive overview of the defence and security issues and policies for different countries covering themes such as the demand and supply of defence, the defence industry, defence and security policy etc. In volumes 11–20 the series on country surveys continued to receive a moderate input of papers. Compared with the first ten volumes, a marginally lower number of surveys were published: nine in total compared with 11 in the first ten volumes (Tables I and II). The list of the country surveys published in last ten volumes is shown in Table IV. It includes a survey of a former super power the Soviet Union and its successor, Russia; the two Koreas and Japan; South Africa, Mexico, Uruguay and New Zealand. Only one NATO member was surveyed in the last ten years, Belgium; and one EU country, Cyprus. In total, 21 surveys have been published thus far, if the UK survey already published in volume 21 is included. The average is one survey per volume during the 20 years of DPE's publication. There is clearly scope for more country surveys since some important countries are missing from our collection. Examples include countries such as the USA, China, Taiwan, Brazil, Egypt, Jordan, Argentina, Sweden, Austria, the Netherlands and many more. More country surveys have been promised and are expected for major countries such as Italy and France. In our view, there is the need to solicit more country surveys as

TABLE IV Country Surveys Volumes 11–20

Country Surveyed	*Author(s)*	*Volume (Issue)*
South Africa	André Roux	11(1) 2000
Japan	Michael Chinworth	11(2) 2000
Cyprus	Christos Kollias	12(6) 2001
Belgium	Wally Struys	13(1) 2002
Soviet Union and Russia	Christopher Davis	13(3) 2002
New Zealand	Robert Alexander and Alan King	13(4) 2002
North and South Korea	Jun Sik Bae	14(1) 2003
Mexico	Eduardo Morales Ramos	15(1) 2004
Uruguay	Amílcar Andrés Peláez	18(3) 2007

Countries surveyed in vols 1–10: India, Sri Lanka, West Germany, Pakistan, Poland, Australia, Greece, Yugoslavia, Spain, Turkey, Israel

well as papers that take a fresh look at countries already surveyed and how and to what extent their defence and security policies have been affected by developments and changes on the regional as well as the global security environment. We would welcome inquiries from interested authors.

A second distinctive characteristic of DPE is that of special issues. In volumes 11–20 there were 20 special issues. Again, on average, one special issue was produced per volume, with a wide spectrum of topics. Thanks to the efforts of the various Guest Editors, DPE has published special issues on topics such as arms exports, civil wars, the fiscal impact of defence

TABLE V Special Issues: Topics and Guest Editors

Topic		*Guest editor(s)*	*Volume (Issue)*
1	Arms exports, controls and production	Paul Levine; Somnath Sen; Ron Smith	11(3) 2000
2	Defence, peace and economic growth in South Africa	André Roux	11(4) 2000
3	Economic aspects of defence in Greece and Turkey	Selami Sezgin; Keith Hartley	12(1) 2001
4	The economics of defence: perspectives from Spain	Alain Cuenca; Aurelia Valiño	12(5) 2001
5	The economics of military affairs in Greece and Turkey	Julide Yildirim; Selami Sezgin	13(2) 2002
6	Military expenditure and development: crowding-out and the effects of military tension and conflict	Christos Kollias	13(5) 2002
7	Economic analysis of civil wars	Todd Sandler; Håvard Herge	13(6) 2002
8	Symposium on defence economics	Julide Yildirim; Selami Sezgin	14(2) 2003
9	Historical perspectives of defence and peace economics	Jurgen Brauer	14(3) 2003
10	Internal and external threats: defence economic analysis	Carlos Pestana Barros; Todd Sandler	14(6) 2003
11	Deficits, debt and defence	Gülay Günlük-Senesen	15(2) 2004
12	The economics of defence: perspectives from Canada	Ugurhan Berkok; Binyam Solomon	16(3) 2005
13	Security challenges and threats in a post-9/11 world	Carlos Pestana Barros; Christos Kollias; Todd Sandler	16(5) 2005
14	Defence research and development in the 21st century	Oren Setter; Asher Tishler	17(3) 2006
15	Canadian perspectives on peacekeeping	Ben Solomon; Ugurhan Berkok	17(5) 2006
16	Memorial essays in honour of Jack Hirshleifer	Martin C. Mcguire	17(6) 2006
17	The economic and political aspects of state terrorism, defence spending, wars and international conflicts	Eftychia Nikolaidou	18(5) 2007
18	Defence economics: perspectives from Turkey	Onur Özsoy	19(3) 2008
19	Defence spending: determinants, economic impact and burden sharing issues	Eftychia Nikolaidou	19(4) 2008
20	War and peace issues in the history of economic thought	Fanny Coulomb; Renaud Bellais	19(5) 2008

expenditures, war and peace issues in the history of economic thought or special issues with a country-specific focus, such as Canada, Greece, Turkey, Spain, South Africa (Table V).

Finally, to conclude this section, a list of the authors of the papers per thematic group for the last ten volumes of the journal is presented in Table VI. As pointed out earlier, we had to exercise our judgment and, as a result of this subjective decision, it is quite possible that some papers were not placed in the category that their authors would have regarded as more fitting and appropriate. We can only apologise for this.

TABLE VI Authors in Volumes 11–20

Defence and the economy (56)	
Peter Batchelor; J. Paul Dunne; David S. Saal	Christos Kollias; Stelios Makrydakis
Paul Dunne; Eftychia Nikolaidou; André Roux	Paul Dunne; Eftychia Nikolaidou; Dimitrios Vougas
Alvin Birdi; Paul Dunne; David S. Saal	Paul Dunne; Eftychia Nikolaidou
David Lai	Selami Sezgin
Michael P. Gerace	James P. Scott
Sonmez Atesoglu (2)	Emmanuel Athanassiou; Christos Kollias; Stavros Zografakis
Ching-chong Lai; Jhy-yuan Shieh; Wen-Ya Chang	Yousif Khalifa Al-Yousif
Carlos Barros	Hannah Galvin
Emmanuel Athanassiou; Christos Kollias	Thanasis Maniatis; Christos Kollias
Jesús Crespo Cuaresma; Gerhard Reitschuler	Selami Sezgin; Jülide Yildirim
Hasan Ersel	Thilo Klein
Erdal Karagol; Serap Palaz	Christos Kollias; Charis Naxakis; Leonidas Zarangas
Chung-Nang Lai; Bwo-Nung Huang; Chin-Wei Yang	Jr-Tsung Huang; An-Pang Kao
Jülide Yildirim; Selami Sezgin; Nadir Öcal	Paul Dunne; Duncan Watson
Paul Dunne; Ron Smith; Dirk Willenbockel	Meenakshi Rishi; Trish Kelly
Selami Sezgin	Anthony J. W. Turner
Binyam Solomon	Hamid E. Ali
Joanne Evans; Eleftherios Goulas; Paul Levine	Oya S. Erdogdu
Jhy-Yuan Shieh; Wen-Ya Chang; Ching-Chong Lai	Albert Wijeweera; Matthew J. Webb
Luca Pieroni	Jenn-Hong Tang; Cheng-Chung Lai; Eric S. Lin
Stella Karagianni; Maria Pempetzoglu	Shin-Jen Tzeng; Ching-Chong Lai; Chun-Chieh Huang
Durmuş Özdemir; Ali Bayar	Nikolaos Mylonidis
Onur Özsoy	Dimitris Hatzinikolaou
Jeffrey S. Smith; M. H. Tuttle	Chien-Chiang Lee; Sheng-Tung Chen
Christos Kollias; Nikolaos Mylonidis; Suzanna-Maria Paleologou (2)	Paresh Kumar Narayan; Baljeet Singh
Neal Wagner; Jurgen Brauer	Rabia Aslam
Chien-Chiang Lee; Sheng-Tung Chen	Chien-Chiang Lee; Chun-Ping Chang
Jülide Yildirim; Nadir Öcal	
Terrorism (37)	
Carlos Barros	James Yetman
Dirk T. G. Rúbbelke	Tony Addison; Mansoob Murshed
João Ricardo Faria; Daniel Arce	Charles H. Anderton; John R. Carter
Raphaël Franck; Arye L. Hillman; Miriam Krausz	Peter J. Phillips
Peter Hilsenrath	Jhy-Yuan Shieh; Jhy-Hwa Chen; Juin-Jen Chang; Ching-Chong Lai
John L. Scott	Tilman Brück
Ziv Naor	Anton D. Lowenberg; Timothy Mathews
Gil S. Epstein; Ira N. Gang	Mete Feridun; Selami Sezgin
Bruno S. Frey; Dominic Rohner (2)	Mehmet E. Yaya
Jason Briggeman; Jeremy Horpedahl	Peter J. Phillips
Carlos P. Barros; Guglielmo Maria Caporale; Luis A. Gil-Alana	Beom S. Lee; Walter Enders; Todd Sandler
Walter Enders; Yu Liu; Ruxandra Prodan	Bahar Araz-Takay; K. Peren Arin; Tolga Omay
Kevin Siqueira; Todd Sandler	Rafael Llorca-Vivero
Carlos Pestana Barros; Isabel Proença; João Ricardo Faria; Luis A. Gil-Alana	Grace F. Sanico; Makoto Kakinaka
Claude Berrebi; Darius Lakdawalla	Emizet Kisangani; E. Wayne Nafziger
David K. Levine; Robert A. Levine	Ian Bellany
Karen Pittel; Dirk T. G. Rübbelke	Chien-Chiang Lee; Chun-Ping Chang
Carlos Pestana Barros; Luis A. Gil-Alana	Ana Bela Santos Bravo; Carlos Manuel Mendes Dias
	Konstantinos Drakos; Andreas Gofas

TABLE VI *(Continued)*

Defence policies, planning & management (30)	
Laura Baldwin; Ellen Pint	Victoria Greenfield; David Persselin
P. G. Pugh	Laura H. Baldwin; Glenn A. Gotz
Maria Hedvall	Eric J. Fredland
David Kirkpatrick	Edward G. Keating; Matthew Dixon
W. J. Hurley	Chad D. Meyerhoefer; Robert P. Trost
David Kirkpatrick	Jonathan Lipow; Eli Feinerman
Aurelia Valiño Castro	Arthur Grimes; James Rolfe
Ralph Rotte; Christoph Schmidt	Richard Fullerton
Vincent Medina; Cyr-Denis Nidier	J. Craig Stone; Binyam Solomon
Philippe Lagassé	Ugurhan G. Berkok
J. Craig Stone	Fanny Coulomb; Jacques Fontanel
Michael A. Rostek	Jeffrey S. Smith; Michael McKee
Shirley J. Ho	Mark Pyman; Regina Wilson; Dominic Scott
Jonathan Lipow; Yakir Plessner	Oren Setter; Asher Tishler
Kazunori Tanigaki	
Conflict and civil wars (29)	
Charles H. Anderton (2)	Heejoon Kang; Rafael Reuveny
Anthony Lyons	Paul Collier; Anke Hoeffler
Nicholas Sambanis	Kevin Siqueira
James Murdoch; Todd Sandler	Jean-Paul Azam; Tahsin Saadi-Sedik
Halvor Mehlum; Karl Moene	Robert Rider
Marta Reynal-Querol	Martin C. McGuire
Gordon Tullock	Sacit Hadi Akdede; Ayla Oğuş
Andrew M. Francis	Peter Sandholt Jensen; Kristian Skrede Gleditsch
Yang-Ming Chang; Shane Sanders	Amnon Levy
Sacit Hadi Akdede; Jinyoung Hwang; Emre Can	Patrick T. Brandt; T. David Mason; Mehmet Gurses; Nico-
Steve R. Garrison	lai Petrovsky; Dagmar Radin
Sang Hoo Bae; Attiat F. Ott	Ibrahim Elbadawi; Håvard Hegre
Solomon Polachek; Carlos Seiglie; Jun Xiang	Amnon Levy; João Ricardo Faria
Malcolm Chalmers	Yang-Ming Chang; Joel Potter; Shane Sanders
Stergios Skaperdas	
Defence industries & industrial base (25)	
Alfonso A. Diaz; Emily O. Goldman	Tim Williams
John S. Earle; Ivan Komarov	Todd A. Watkins; Maryellen R. Kelley
Derrick J. Neal; Trevor Taylor	Carlos Barros
Benny Mantin; Asher Tishler	Christos Kollias; Apostolos Rafailidis
Eric Gates	Ian Jackson
Paul Dowdall; Derek Braddon; Keith Hartley	Paul Dowdall
Keith Hayward	Wally Struys
Yoad Shefi; Asher Tishler	Damien Besancenot; Radu Vranceanu
Lawrence McDonouch	Peter Batchelor; Paul Dunne; Sepideh Parsa
Philip G. Pugh	Kogi Balakrishnan; Ron Matthews
Cedric Laguerre	Mark van de Vijver; Bart Vos
Keith Hartley	
Paul Dunne; María del Carmen García-Alonso;	
Paul Levine; Ron Smith	
Peace and peacekeeping (23)	
Hirofumi Shimizu	John R. Oneal; Bruce Russett
Patrick James; Eric Solberg; Murray Wolfson	Jurgen Brauer; André Roux
Peter Stauvermann	Martin C. McGuire
Dietrich Fischer; Jurgen Brauer	Benno Torgler
Emmanuel Athanassiou	Kjell Hausken
James K. Galbraith; Corwin Priest; George Purcell	Jacques Fontanel; Jean-Paul Hebert; Ivan Samson
Jacques Fontanel; Fanny Coulomb	Fanny Coulomb; Renaud Bellais
Jacques Fontanel; Liliane Bensahel; Steven Coissard;	Renaud Bellais; Fanny Coulomb
Yann Echinard	Charles Wolf Jr
Fanny Coulomb; Keith Hartley; Michael Intriligator	Ugurhan G. Berkok
Ross Fetterly (2)	Dane Rowlands; David Carment
Louis Parai	

TABLE VI *(Continued)*

Arms races, arms control and disarmament (22)	
Ron Smith; Paul Dunne; Eftychia Nikolaidou	Michael D. Intriljgator; Dagobert L. Brito
Jacques Fontanel; Fanny Coulomb	Cassady B. Craft
Christos Kollias; Kleanthis Sirakoulis	Christos Kollias; Suzanna-Maria Paleologou
A. S. Andreou; G. A. Zombanakis	A. S. Andreou; K. E. Parsopoulos; M. N. Vrahatis; G. A.
Nadir Öcal (2)	Zombanakis
Fanny Coulomb; Jacques Fontanel	Toshihiro Ihori
A. S. Andreou; N. H. Mateou; G. A. Zombanakis	Minoru Okamura
Vasilis Zervos	Kobi Kagan; Asher Tishler; Avi Weiss(2)
Jun Sik Bae	A. S. Andreou; G. A. Zombanakis
Martin C. Mcguire	Nadir Öcal; Jülide Yildirim
Hasan Şahin; Onur Özsoy	
Economics of alliances (20)	
Minoru Okamura	Kevin Siqueira; Todd Sandler
Pedro Gonzalez De La Fe; Daniel Montolio	William Gates; Katsuaki Terasawa
Claudia Pérez-Formés; Alain Cuenca	Dolores Gadea; Eva Pardos; Claudia Pérez-Forniés
Binyam Solomon (2)	A. S. Andreou; K. E. Parsopoulos; M. N. Vrahatis; G. A.
Todd Sandler (2)	Zombanakis
Marc Guyot; Radu Vranceanu	Bruce G. Linster; Richard L. Fullerton; Michael Mckee;
Keith Hartley	Stephen Slate
Loek Groot; Vincent van den Berg	Karl Derouen Jr; Uk Heo
Jomana Amara (2)	Christos Kollias
Prabal Roy Chowdhury	
Defence R&D and technology (17)	
Jordi Molas-Gallart	Mikel Buesa
David S. Saal	Richard B. Streeter; Guy W. Hagen; Edward E. Patenaude;
Eduardo Morales-Ramos	Dennis K. Killinger
David Kirkpatrick	Todd A. Watkins
Keith Hartley	R. Wylie; S. Markowski; P. Hall
Manuel Trajtenberg	Renaud Bellais; Renelle Guichard
Oren Setter; Asher Tishler	David W. Versailles; Valérie Mérindol
Andrew D. James	Mark Van de Vijver; Bart Vos
Andrew Middleton; Steven Bowns; Keith Hartley;	Vasilis Zervos; G.M. Peter Swann
James Reid	
Fiscal effects of defence spending (16)	
Randa Alami	Onur Ozsoy
Gülay Günlük-Senesen (2)	Paul Dunne; Sam Perlo-Freeman; Aylin Soydan (2)
Jon Jonakin; Mark Stephens	Christos Kollias; George Manolas; Suzanna-Maria
Selami Sezgin	Paleologou
Erdal Karagol (2)	Erdal Karagol; Selami Sezgin
Yemane Wolde-Rufael	Erdal T. Karagol; Aziz Turhan
Paresh Kumar Narayan; Seema Narayan	Onur Özsoy
Military personnel & recruitment (14)	
John T. Warner; Beth J. Asch	Andy Thorpe; Sam Cameron
Alexei Izyumov; John Vahaly	Ian Bellany
James R. Hosek; Mark E. Totten	Michael L. Hansen; Jennie W. Wenger
John T. Warner; Sebastian Negrusa	John Warner; Curtis Simon; Deborah Payne
Katarina Keller; Panu Poutvaara; Andreas Wagener	Curtis J. Simon; John T. Warner
Peter Macdonald	Shabbar Jaffry; Yaseen Ghulam; Alexandros Apostolakis
Jülide Yildirim; Bülent Erdinç	
Models of defence expenditure (12)	
David Throsby; Glenn A. Withers	Selami Sezgin; Jülide Yildirim
Christos Kollias; Suzanna-Maria Paleologou	Paul Dunne; Eftychia Nikolaidou; Nikolaos Mylonidis
Paul Dunne; Sam Perlo-Freeman	Binyam Solomon
Gülay Günlük-Şenesen	María-Dolores Gadea; Antonio Montañes
John Hudson; Philip Jones	Thomas Bernauer; Vally Koubi; Fabio Ernst
J. Paul Dunne; Sam Perlo-Freeman; Ron P. Smith	Eftychia Nikolaidou

TABLE VI *(Continued)*

Arms trade (10)	
Keith Hartley	Paul Levine; Fotis Mouzakis; Ron Smith
Jurgen Brauer	Todd Sandler
María del Carmen García-Alonso; Keith Hartley	Cassady Craft; Joseph Smaldone
Michael Brzoska	G. Sharp
Stefan Markowski; Stephanie Koorey; Peter Hall; Jurgen Brauer	Pavel Yakovlev
Country surveys (9)	
André Roux	Michael W. Chinworth
Christos Kollias	Wally Struys
Christopher Davis	J. Alexander; Alan B. King; W. Robert
Jun Sik Bae	Eduardo Morales Ramos
Amílcar Andrés Peláez	
Regional economics and defence (4)	
Andy Pike; James Cornford; John Tomaney	Michael Greenberg; Michael Frisch; Tyler Miller; David Lewis
Michael Asteris; Jeffery Grainger; David Clark; Shabbar Jaffry	Linda Andersson; Johan Lundberg; Magnus Sjöström
Others (7)	
Keith Hartley; Todd Sandler	Jurgen Brauer
Martin C. McGuire	Mansoob Murshed
Hubert Van Tuyll; Jurgen Brauer	Martin C. McGuire; Gary S. Becker
Kjell Hausken	
Book reviews (14)	
Stephen Martin	Nick Hooper; Stephen Martin
Peter Lock	Derek Braddon
Peter Almquist	Carlos Pestana Barros (2)
Keith Hartley (2)	Jonathan Lipow
Atin Basuchoudhary	Ian Jackson
Ron Matthews	Gülay Günlük-Şenesen

Note: Numbers in brackets indicate more than one contribution by the author(s)

GAPS AND FUTURE DIRECTIONS

In this section, we highlight possible gaps in DPEs coverage and even perhaps point to a future research agenda in the context of the evolving international security environment. Of course, such an exercise can prove to be fairly subjective. Our intention is not to narrow future research directions but rather to contribute to the discussion within our academic community.

As already pointed out, the coverage of DPE is directly affected by the changes in the international security environment. If we attempt to summarise periods in the focus of the coverage in DPE then it is possible, in very broad terms, to identify three periods with the corresponding shift in focus: Early post-Cold War, Late post-Cold War and the post 9/11 period. Thus, for example, we have moved from issues of disarmament and conversion in the early post-Cold War years to issues of conventional and non-conventional conflict in last decade, where themes such as terrorism, intrastate and interstate conflict and war emerged as new dynamic research areas. Although the coverage in DPE has followed development in the international scene, one can nevertheless identify some gaps or under-represented fields in the coverage. For instance: the regional effects of defence; cyber security; defence budget trade-offs; (counter)insurgency; joint international defence industrial projects; the direct and indirect costs of war and conflict; the economic dimension of common European defence policy;

international collective action vis-à-vis the new global security challenges; defence and the environment; genocide; experimental analyses and many more. Economic analysis offers many useful tools and methodologies with which such topics can be approached, examined and analysed.

Apart from the gaps in the coverage – and these are not, by any means, the only ones – we should attempt to take a look to the next ten years. Currently, it seems that they will be characterized by the following parameters that will probably shape the defence and security environment: the 'globalization' of security challenges, referring to issues associated with the continued threat of terrorism, rogue states, intrastate conflict, etc. At the same time, given the fiscal problems that many countries in Europe and elsewhere face, shrinking defence or security budgets will be another parameter that will shape the years to come. This will give rise to discussions and debates around the defence and security policy choices and options available to meet the challenges of the international environment. More specifically, in recent years, the concept of national security has profoundly changed. In the Cold War period, it was associated with border defence, vis-à-vis a rival state or states and the threat they represented to national security interests. In the post bipolar world, and especially in the post 9/11 period, the concept of national security has broadened to include threats such as transnational terrorism or even international organized crime. If this is the case, the following implications emerge. Military expenditure is no longer the only variable that encapsulates the costs of producing security. In other words, the budgets of other security forces are becoming important in calculating the costs of producing security. If this is accepted, then the following issues emerge. How broadly should we define national security and hence what institutions should we include in calculating the costs of national security spending? Do we have comparable international data? Assuming we have the data, what are the economic, fiscal and other effects of this security expenditure? To the best of our knowledge, comparatively very little research has thus far been undertaken on the issue of public security spending in most countries, perhaps with the exception of the US. Furthermore, if security is defined in such broader terms then this has implications when we come to examine the industry that produces security inputs. Clearly, this is no longer limited to the traditional defence industry but should be expanded to include the many firms engaged in producing security services and products and organizations such as INTERPOL. It would appear that this broader definition of national security opens up a host of issues that in the future may attract the focus of the research agendas in our academic community. Similarly, the rapidly unfolding environment of fiscal constraints and shrinking defence budgets also creates interesting areas for research. For instance: the choices and possible trade-offs within defence budgets; choosing what capabilities to keep and what to forego; how to improve efficiency via, for example, competitive outsourcing in the provision of military services; specialization and pooling between nations for example in the context of a common European defence policy; ad-hoc alliances to counter specific threats or to achieve specific targets; alternative means of achieving protection or costs and benefits of alternative policies vis-à-vis the final outputs in terms of defence, security and peace. All these are areas where the tools of economic analysis can offer useful insights and, hopefully, in the next ten years we will see many interesting papers published on such topics in DPE.

CONCLUDING REMARKS

Before this summary of the second decade of DPE is concluded, we feel that we need to gratefully acknowledge the invaluable contribution that both the founding editors, as well all those who have served either as editors or members of the editorial board, have made over the years

towards its success. Last but not least, we want to sincerely thank all the reviewers that have supported the journal with their timely, constructive and valuable input. They have ensured the quality of the published papers and hence the journal's standing.

References

Bruck, T. (2005) An economic analysis of security policies. *Defence and Peace Economics* **16**(5) 375–389.
Enders, W. and Sandler, T. (2006) *The Political Economy of Terrorism.* Cambridge: Cambridge University Press.
Hartley, K. (2006) Defence economics: achievements and challenges. Mimeograph.
Hartley, K. and T. Sandler (2000) *Defence and Peace Economics:* a ten-year retrospective. *Defence and Peace Economics* **11**(1) 1–16.
Hitch, C. and R. McKean (1960) *The Economics of Defence in the Nuclear Age.* Harvard University Press.